

Feature Selection

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Introduction: Feature

TRS_DT	TRS_TYP_CD	REF_DT	REF_NUM	CO_CD	GDS_CD	QTY	UT_CD	UT_PRICE
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21/05/93	00001	14/05/93	25232	10026H	006C	10	CTN	69.000
21/05/93	00001	14/05/93	25235	10027E	003A	5	CTN	24.000
21/05/93	00001	14/05/93	25235	10027E	001M	5	CTN	24.000
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21/05/93	00001	27/04/93	25033	10035E	015A	375	GRS	72.000
21/05/93	00001	20/05/93	25313	10041Q	010F	10	CTN	26.000
21/05/93	00001	12/05/93	25197	10054R	002E	25	CTN	24.000

Introduction: Feature Selection Methods

- Feature selection methods since the 1970's.
- Common steps in feature selection
- Key concepts in feature selection
- Categorization of Feature Selection algorithms.
- Pros and Cons of Feature Selection algorithms.

Feature Selection for Classification

- What is Classification?
 - Why supervised in nature?
 - Importance of Class label
 - Prediction of class label of a new instance based on prior knowledge using appropriate logic.
 - Strength and weaknesses

Feature Selection for Classification

- Common Scenario:
 - Given a large real-life data set.
 - Given a number of known classes.
 - Select an appropriate number of training instances.
 - Build a Prediction Model (eg. Decision tree).
 - Use the model to classify the test instances into the pre-defined classes.
 - Evaluate the performance using appropriate statistical measures (such a precision, recall, ROC, AUC, etc).

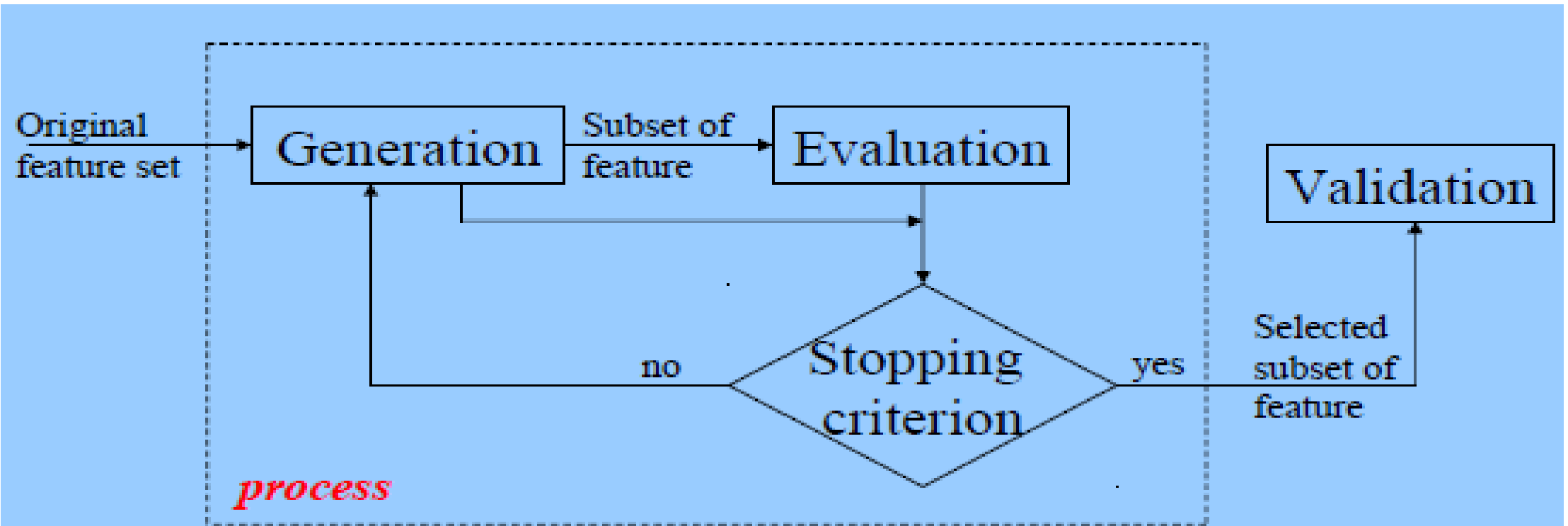
Feature Selection for Classification

- Study the abilities of various known feature selection methods experimentally using real-life/ benchmark datasets.
- Consider the subsets of relevant features given by a method
- Use the subset of features to classify test instances and justify the effectiveness of the method in terms of statistical measures.
- Higher the **Relevancy**, better will be the **Prediction Performance**.

Why not Full Feature Set?

- In some situations (datasets with large dimensionality), too expensive from both computationally as well as resource consumption perspective to examine all features.
- All features are usually not relevant, some are redundant.
- Irrelevant features can degrade the classification performance.

Feature Selection: Major Steps



Generation

= select feature subset candidate.

Evaluation

= compute relevancy value of the subset.

Stopping criterion = determine whether subset is relevant.

Validation

= verify subset validity.

Feature Selection Steps : Subset Generation

- Select candidate subset of feature for evaluation. Start with-
 - **Case 1**: No feature,
 - **Case 2**: All features,
 - **Case 3**: Random feature subset.
- Subsequently-
 - Add features (**Case 1**),
 - Remove features (**Case 2**),
 - Add or Remove (**Case 3**).
- Categorize feature selection methods based on 'ways to generate candidate feature subsets'.
 - Complete
 - Heuristic
 - Random.

Feature Selection Steps : Complete or Exhaustive

- This approach generates all possible combinations of feature subsets.
 - For example, for a given feature set $F=\{f_1, f_2, f_3\}$, it will generate following all possible subsets for examination: $\{\{f_1\}, \{f_2\}, \{f_3\}, \{f_1, f_2\}, \{f_1, f_3\}, \{f_2, f_3\}, \{f_1, f_2, f_3\}\}$
- Although it guarantees for finding an optimal subset of features, the order of the search space $O(2^p)$, where p is the number feature.
- It is too expensive, especially when the cardinality of the feature set is large.

Feature Selection Steps : Heuristic

- Here, selection of subsets is directed by certain pre-specified guidelines.
- As per guideline, some subsets of features are taken out, assuming those subsets will not contribute the classification process.
 - Possible candidate sets = $\{ \{f_1, f_2, f_3\}, \{f_2, f_3\}, \{f_3\} \}$
- Often, this approach exploits an incremental approach for generation of subsets. With reference to a feature (assumed to be most relevant), it grows.
- Search space is smaller and faster in producing result.
- May miss out some important feature subset(s).

Feature Selection Steps : Random

- No predefined way is used to select candidates for a feature subset.
- Attempts to pick a feature at random (a probabilistic approach).
- Optimality in terms of cardinality of the final feature subset depends on the number of iterations used.
- Requires more user-defined input parameters.
- Optimality depends on how these parameters are defined and number of attempts.
- No guarantee of achieving best possible subset of features.

Feature Selection Steps : Evaluation

- For evaluation of a given feature subset, consider
 - Relevance of the subset towards the classification task.
 - Independence among the features in a set.
- Following evaluation functions are commonly used.
 - Distance (Using proximity measures such as Euclidean distance).
 - Information (Using information theoretic measures: entropy, MI, etc.)
 - Dependency (Using Pearson correlation coefficient).
 - Consistency (Using min-features bias).
 - Classifier error rate (Using a classifier).

Feature Selection Steps : Evaluation Functions

(8.1) Distance measure

- $z^2 = x^2 + y^2$
- select those features that support instances of the same class to stay within the same proximity.
- instances of same class should be closer in terms of distance than those from different class.

(8.2) Information measure

- entropy - measurement of information content.
- information gain of a feature : (eg. Induction of decision tree)
 $\text{gain}(A) = I(p,n) - E(A)$
 $\text{gain}(A)$ = before A is branched - sum of all nodes after branched
- select A if $\text{gain}(A) > \text{gain}(B)$.

Feature Selection Steps : Evaluation Functions

(8.4) Consistency measure

- two instances are *inconsistent* if they have *matching feature values* but group under *different class label*.

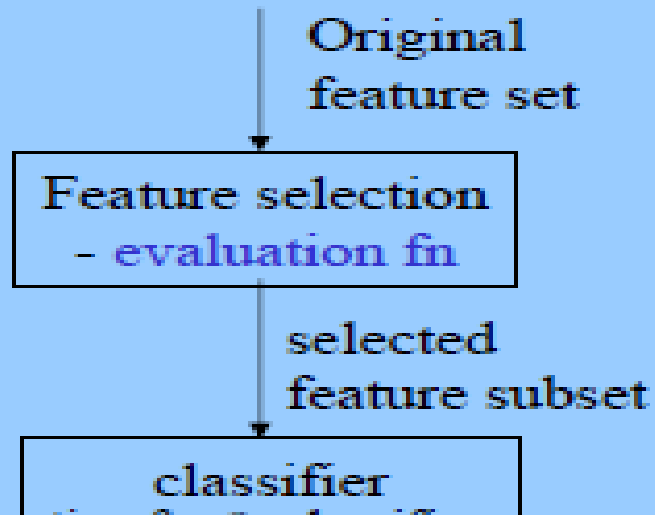
	f_1	f_2	class
instance 1	a	b	c1
instance 2	a	b	c2

← inconsistent

- select $\{f_1, f_2\}$
if in the training data set there exist no instances as above.
- heavily rely on the training data set.
- min-feature = want smallest subset with consistency.
- problem = 1 feature alone guarantee no inconsistency (eg. IC #).

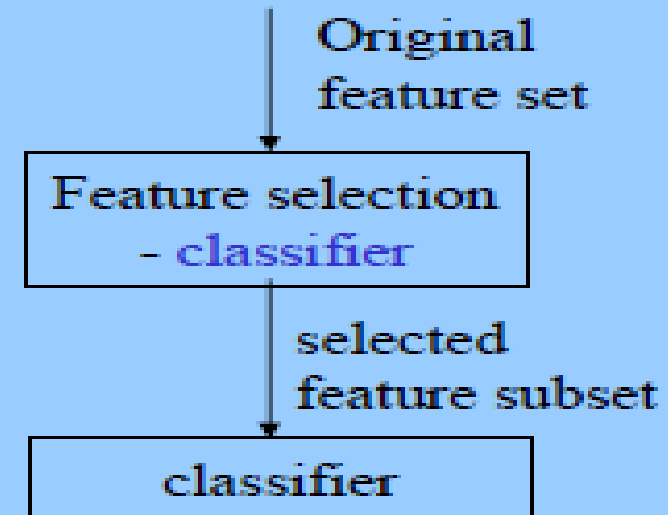
Feature Selection : Filter vs Wrapper

Filter approach



- evaluation fn $\nsubseteq$ classifier
- ignored effect of selected subset on the performance of classifier.

Wrapper approach



- evaluation fn = classifier
- take classifier into account.
- loss generality.
- high degree of accuracy.

Comparison of Evaluation Measures

method	generality	time	accuracy
distance	yes	low	-
information	yes	low	-
dependency	yes	low	-
consistency	yes	moderate	-
classifier error rate	no	high	very high

generality = how general is the method towards diff. classifier?

time = how complex in terms of time?

accuracy = how accurate is the resulting classification task?